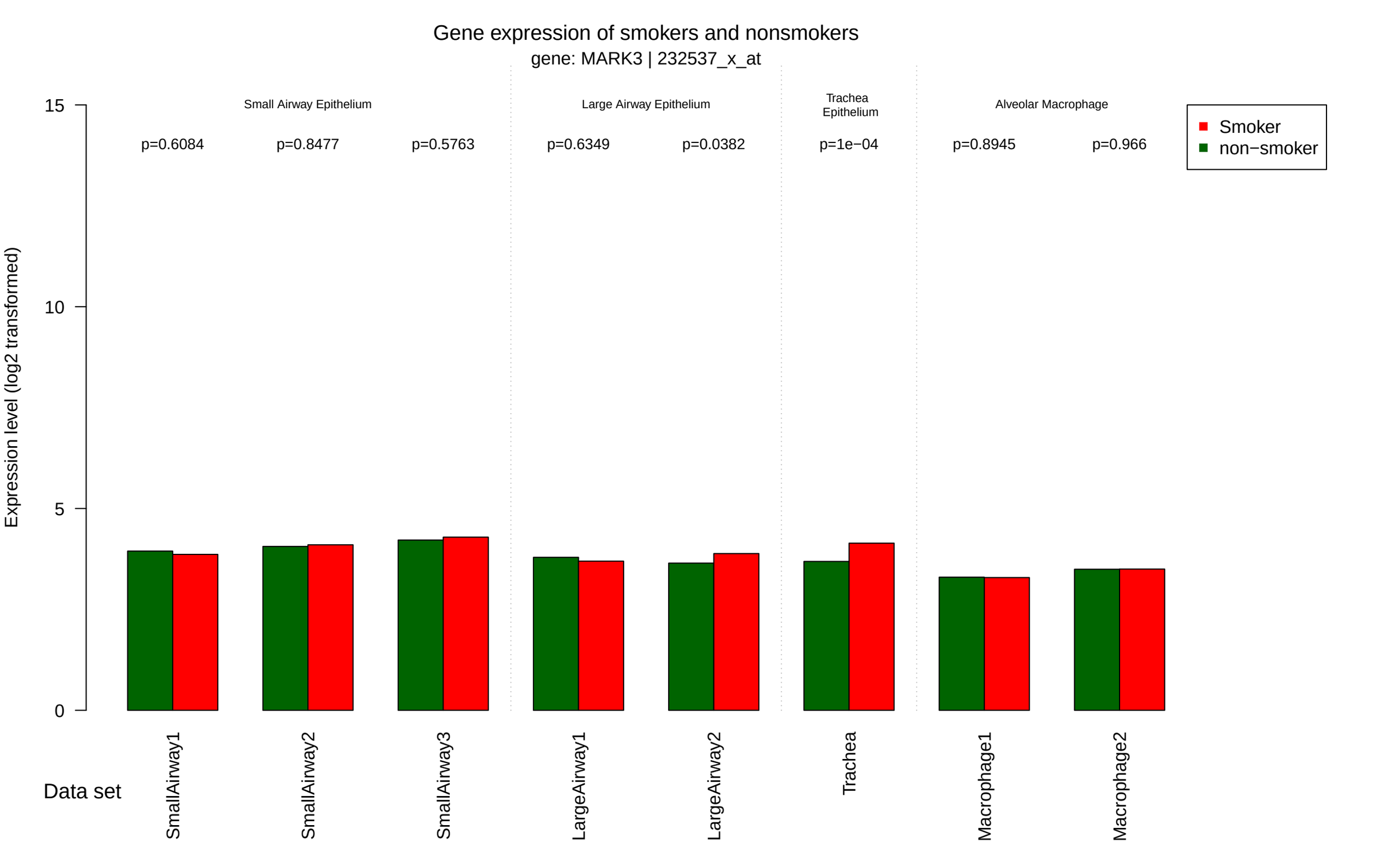
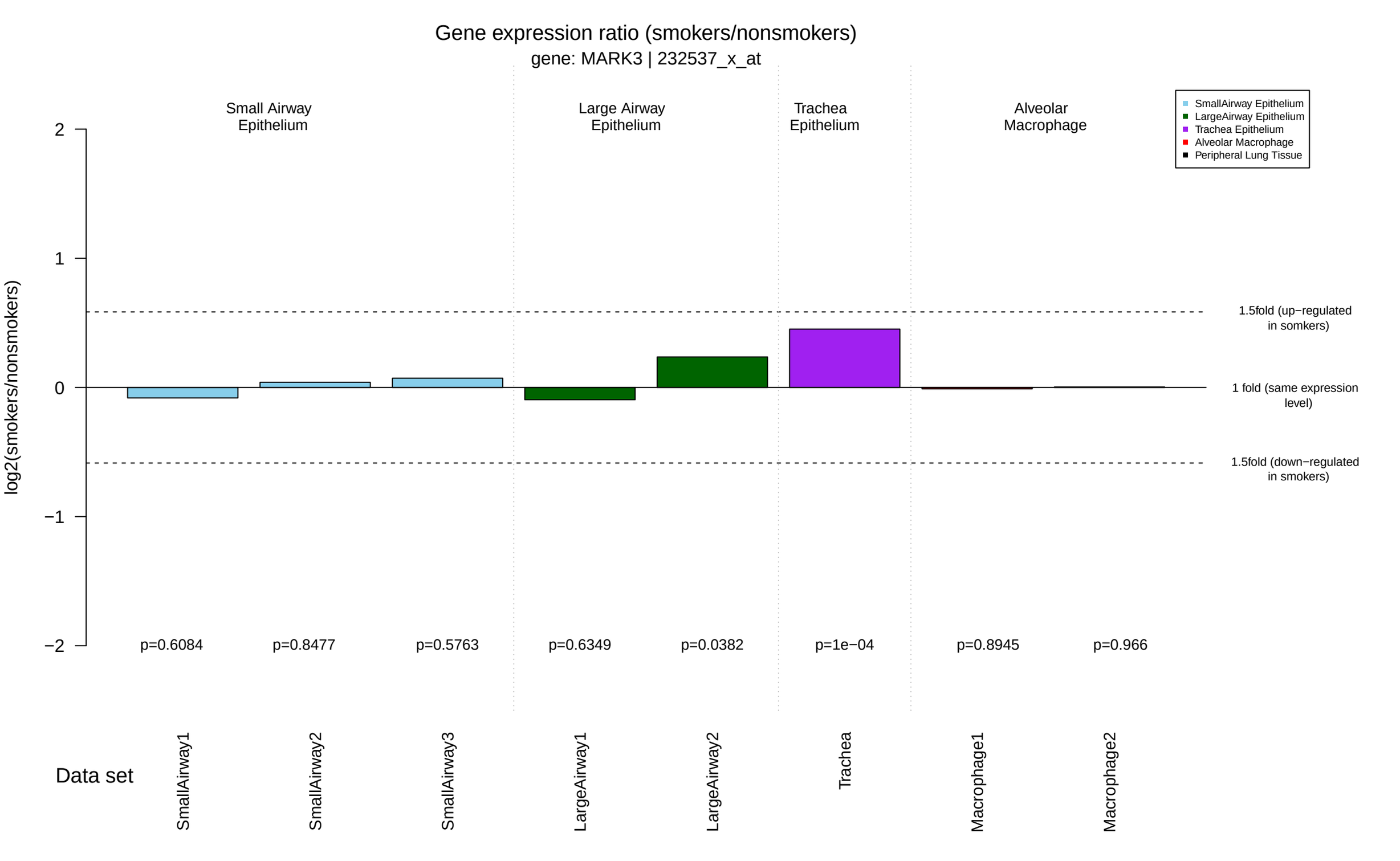
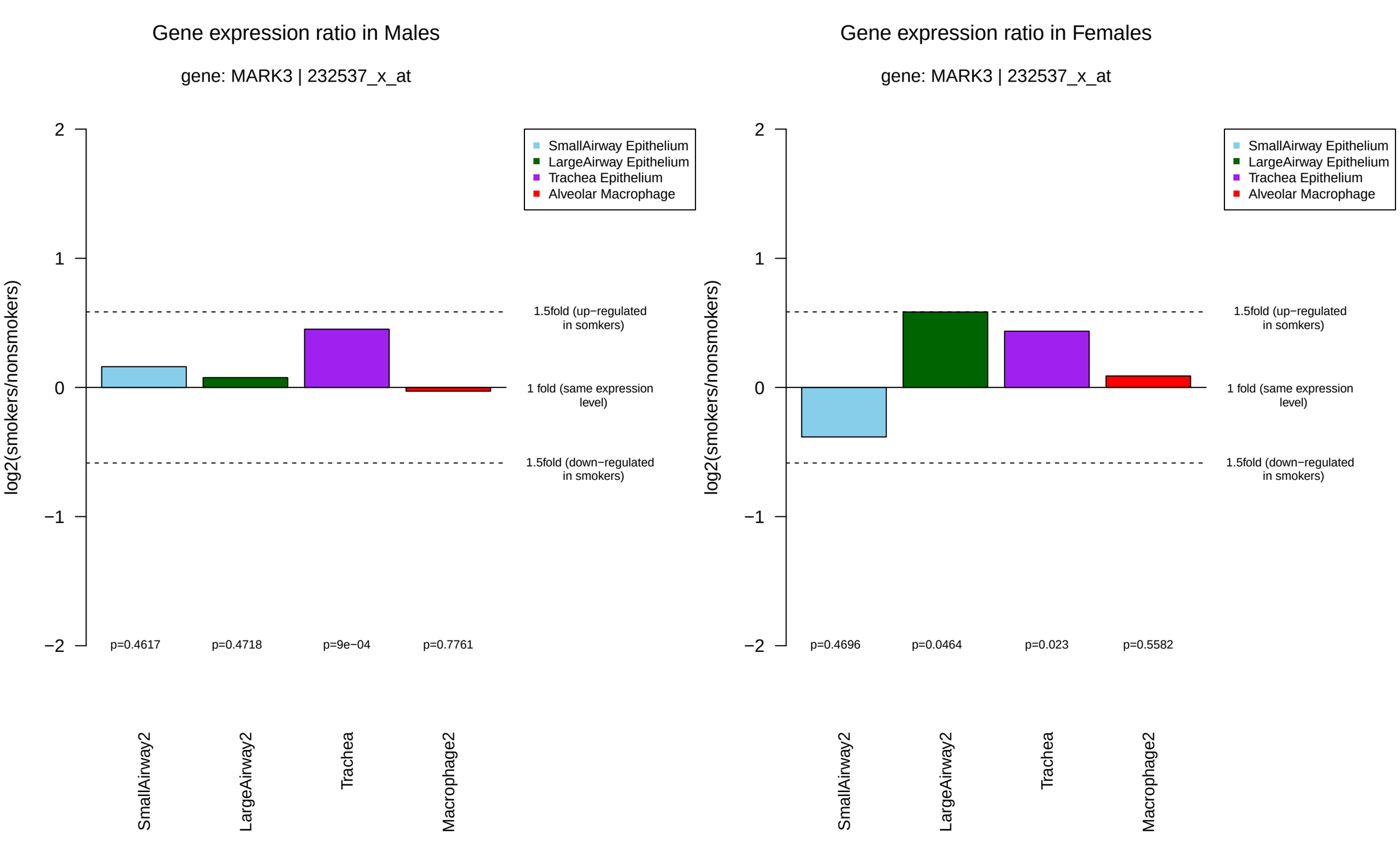
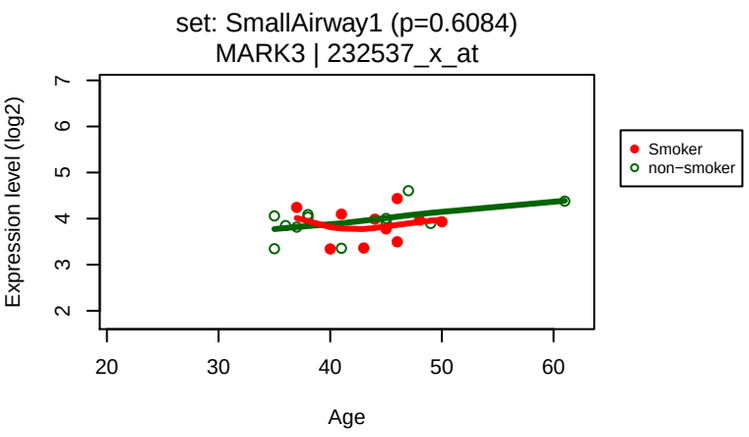
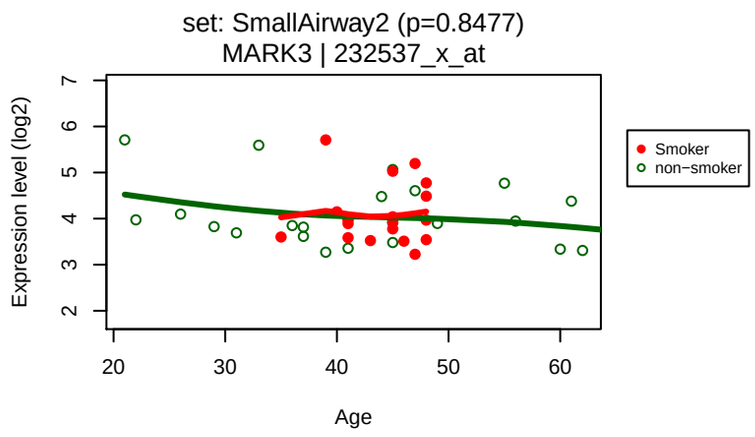
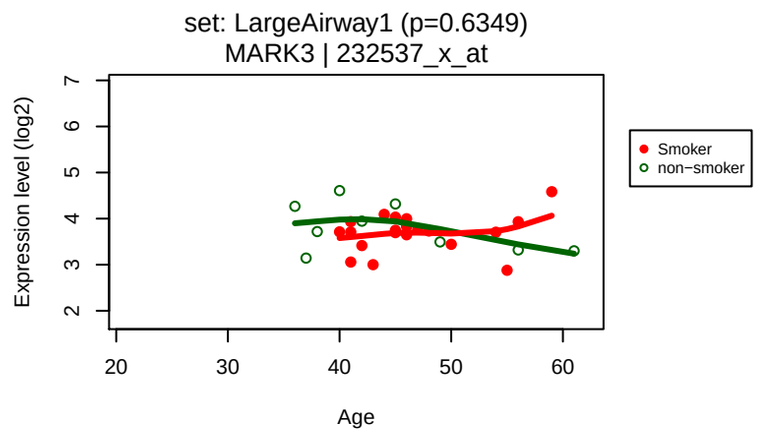

Gene expression of smokers and nonsmokers
gene: MARK3 | 232537_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6084
p=0.8477
p=0.5763
p=0.6349
p=0.0382
p=1e−04
p=0.8945
p=0.966
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

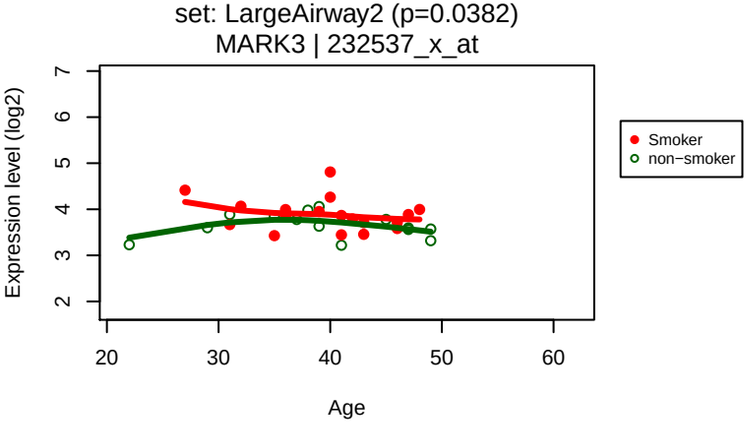

Gene expression ratio (smokers/nonsmokers)
gene: MARK3 | 232537_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6084
p=0.8477
p=0.5763
p=0.6349
p=0.0382
p=1e−04
p=0.8945
p=0.966
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

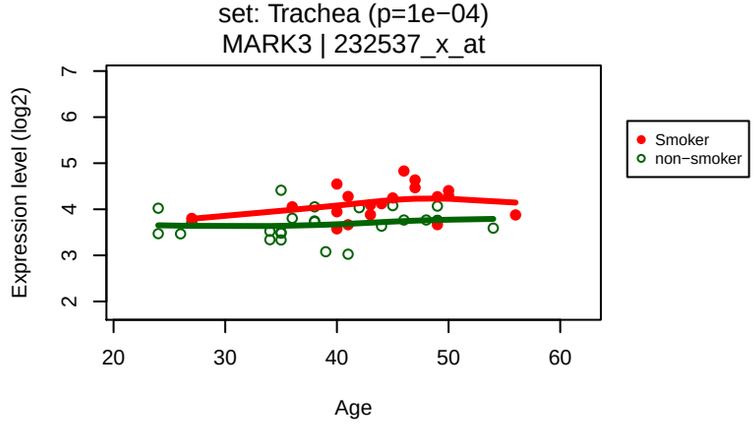

Gene expression ratio in Males
gene: MARK3 | 232537_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4617
p=0.4718
p=9e−04
p=0.7761
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MARK3 | 232537_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4696
p=0.0464
p=0.023
p=0.5582
SmallAirway2
LargeAirway2
Trachea
Macrophage2

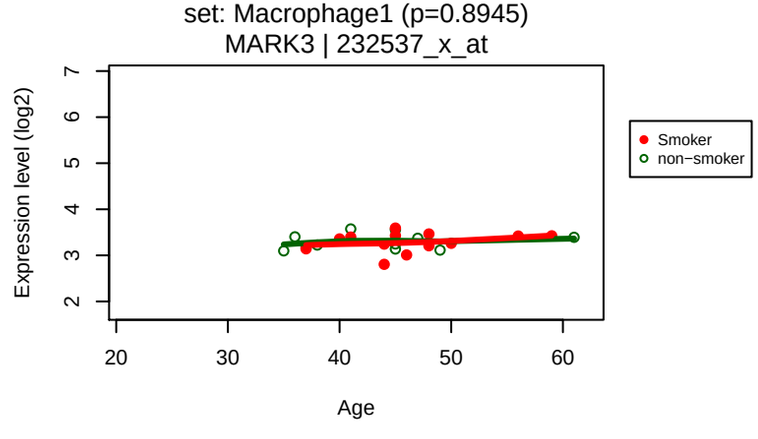

set: SmallAirway1 (p=0.6084)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker

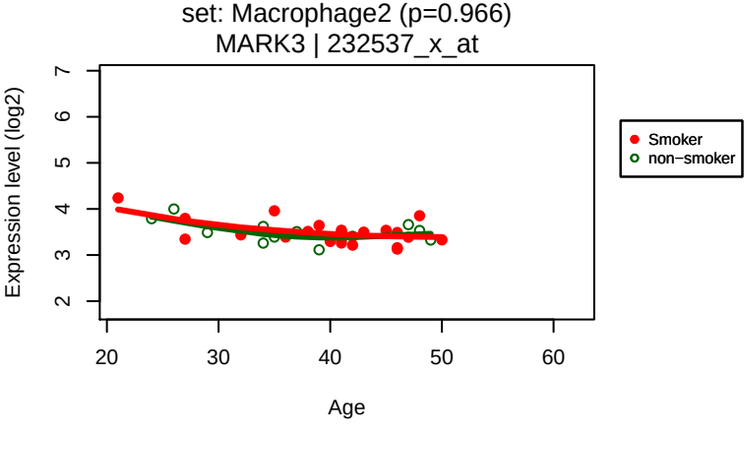

set: SmallAirway2 (p=0.8477)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker

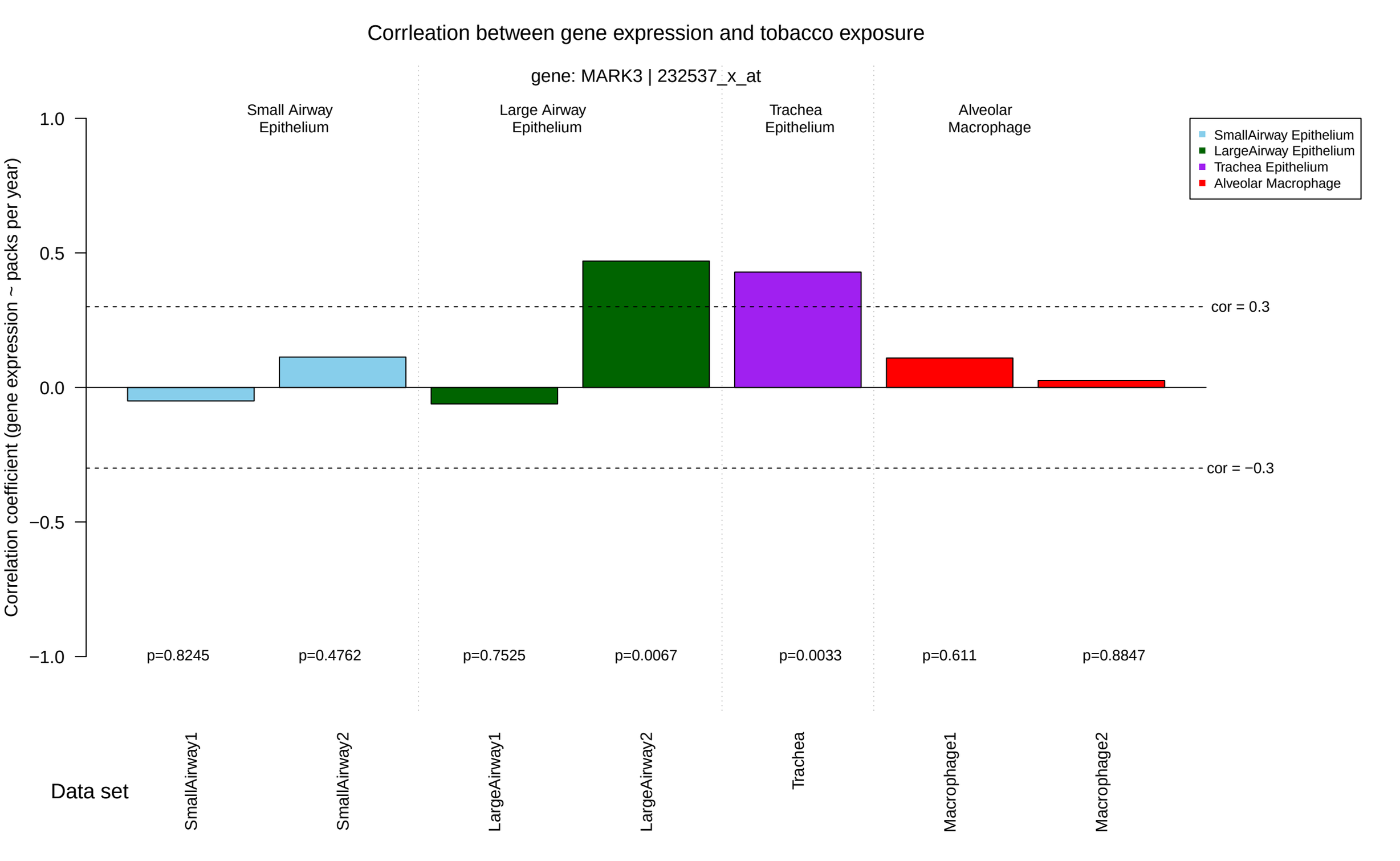

set: LargeAirway1 (p=0.6349)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0382)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8945)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.966)
MARK3 | 232537_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MARK3 | 232537_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8245
p=0.4762
p=0.7525
p=0.0067
p=0.0033
p=0.611
p=0.8847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2